CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 11-2026: A 24-Year-Old Man with Depression, Anhedonia, and Fatigue

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CME

PRESENTATION OF CASE

Dr. Aaron R. Quiggle: A 24-year-old man was evaluated in the outpatient psychiatry clinic of this hospital because of depression, anhedonia, and severe fatigue.

Eighteen months before the current presentation, the patient had been admitted to another hospital for depression, suicidality, and management of opioid withdrawal. He was then transferred to a residential facility for further care, where he received induction treatment with buprenorphine. Treatment with venlafaxine and prazosin was started in the outpatient setting after discharge from the facility.

Four months before the current presentation, the patient presented to the psychiatry clinic of this hospital for treatment of coexisting substance use disorder and mood disorder. He reported long-standing anhedonia, feelings of hopelessness, and depressed mood, which had begun 5 years earlier, when he was a student in high school. The patient also noted severe anxiety with perseveration, especially in social contexts related to how others perceived him, and intrusive thoughts that kept him from falling asleep. Two years before the current presentation, these symptoms had led him to begin using recreational fentanyl intranasally, orally, and occasionally by intravenous injection. In addition, he reported intermittent recreational nonprescribed benzodiazepine use. During this time period, the patient had two unintentional opioid overdoses.

During the initial evaluation at the psychiatry clinic, the patient reported persistent neurovegetative symptoms, including fatigue, anhedonia, anorexia, and low mood. He described poor sleep quality and excessive daytime fatigue, and he stated that he often fell asleep multiple times throughout the day. In addition, he had frequent nightmares, which occurred on most nights. He reported no history of physical or sexual trauma, impulsivity, flight of ideas, delusions, distractibility, obsessive or compulsive behaviors, panic attacks, issues with attention, restlessness, or paranoia.

At a follow-up visit 2 months before the current presentation, the patient reported improved mood and decreased anxiety. However, he also reported substantial and persistent fatigue and daytime somnolence, which led him to fall asleep while on public transportation and in the park while reading. The patient associated these symptoms with the initiation of venlafaxine treatment. At the time of the follow-up visit, he had remained abstinent from opioids and other substances. The dose of venlafaxine was decreased.

On the day of the current presentation, the patient noted anhedonia with a lack of motivation and severe fatigue that limited his ability to complete tasks during the workday. He reported falling asleep several times per day, sometimes for up to 20 hours. He also noted ongoing anxiety, which usually occurred in social situations, and that he sometimes felt weakness in his hands when he laughed, which he attributed to anxiety in social situations. The patient reported that he had had visual and tactile hallucinations while falling asleep, and he also described awakening and feeling paralyzed multiple times per night. He stated that his partner had not told him of any apneic spells or snoring, and he did not have early morning headaches. In addition, the patient said that he had not recently used any opioids other than buprenorphine but indicated that he had begun using ketamine several times per week and drinking two or three alcoholic beverages once per week.

The patient's psychiatric history was notable for opioid use disorder, generalized anxiety disorder, and major depressive disorder. He had no known medical conditions. Medications included buprenorphine, venlafaxine, prazosin, and transdermal and oral nicotine replacement therapy. The patient was in a monogamous relationship, was currently employed, and lived in Boston with his parents and siblings. He had taken a leave of absence from college but was planning to return. His family history was notable for multiple relatives with substance use disorders, as well as reports of unspecified sleep disorders in several first- and second-degree family members. Owing to the ongoing impairment in sleep, venlafaxine was discontinued, and treatment with escitalopram was started.

On examination, the blood pressure was 126/59 mm Hg, the heart rate 81 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. The weight was 77 kg, and the body-mass index (BMI; the weight in kilograms divided by the square of the height in meters) was 21. He was well-appearing, alert, and interactive. The neck circumference appeared to be normal, and the posterior oropharynx was clearly visible without obstruction by the tongue or soft tissue. The neurologic examination was unremarkable, with symmetric cranial-nerve function, a normal gait, and no motor deficits. The speech was fluid, spontaneous, and not pressured. The thought process was linear, and the thought content was devoid of delusions, paranoia, or grandiosity. The patient's attention was preserved. His affect was constricted, but he did not appear to be anxious.

The blood levels of electrolytes, alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, bilirubin, and kidney-function tests were normal. The white-cell count, platelet count, and hemoglobin level were normal. Blood levels of iron, total iron-binding capacity, transferrin, and ferritin were normal. Other laboratory test results are shown in Table 1. Urinalysis was normal, and blood and urine toxicology testing was positive for buprenorphine and norbuprenorphine but negative for other opioids, benzodiazepines, or stimulants. Serologic testing for human immunodeficiency virus, hepatitis B and hepatitis C viruses, and syphilis was negative.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Masoud P. Kamali: The patient presented with anhedonia, depression, anxiety, fatigue, and sleep abnormalities in the context of ongoing polysubstance use disorder. This constellation of symptoms is consistent with an episode of major depressive disorder. The patient started treatment with several medications for these conditions, such as buprenorphine for opioid dependence, venlafaxine for depression and anxiety, and prazosin to mitigate his nightmares. Although the patient's mood improved and anxiety decreased, his daytime somnolence and fatigue persisted, which prompted medication adjustments.

In developing a differential diagnosis to explain the features of this case, I will focus on the most distressing and impairing symptoms reported by the patient — namely, extreme fatigue and intrusive somnolence. Although the severity of his depression and substance use fluctuated over the course of months of treatment, I am

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Current Presentation
Hemoglobin (g/dl)	13.5–17.5	12.6
Hematocrit (%)	41.0–53.0	37.7
White-cell count (per μl)	4500–11,000	5870
Platelet count (per μl)	150,000–400,000	363,000
Sodium (mmol/liter)	135–145	140
Potassium (mmol/liter)	3.4–5.0	4.6
Chloride (mmol/liter)	98–108	106
Carbon dioxide (mmol/liter)	23–32	27
Urea nitrogen (mg/dl)	8–25	19
Creatinine (mg/dl)	0.60–1.50	0.80
Glucose (mg/dl)	70–110	106
Globulin (g/dl)	1.9–4.1	3.8
Albumin (g/dl)	3.3–5.0	4.8
Alanine aminotransferase (U/liter)	10–40	10
Aspartate aminotransferase (U/liter)	10–55	14
Alkaline phosphatase (U/liter)	45–115	63
Total bilirubin (mg/dl)	0.0–1.0	0.3
Iron (μg/dl)	45–160	102
Transferrin saturation (%)	14–50	27
Total iron-binding capacity (μg/dl)	230–404	381
Ferritin (μg/liter)	20–300	40

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for iron and iron-binding capacity to micromoles per liter, multiply by 0.1791.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

most concerned by his sleep-related problems, especially sleeping up to 20 hours a day, falling asleep several times per day, and severe fatigue, all of which limited his ability to complete tasks. These are the core features of excessive daytime sleepiness, and I will review its causes systematically.[1] First, I will review the most common causes of excessive daytime sleepiness in the general population, followed by the most likely causes in this patient on the basis of his coexisting conditions. Then, I will review the rare causes of excessive daytime sleepiness that may be considered if other diagnoses have been ruled out, with a focus on central hypersomnolence disorders.

Common Causes of Excessive Daytime Sleepiness

Excessive daytime sleepiness often occurs in situational states that lead to insufficient sleep, such as those seen in students studying for an exam or in occupational settings such as shift work or jet lag, which impair the sleep–wake cycle. Given that this patient did not report any of these circumstances, they are unlikely causes of his excessive daytime sleepiness.

Obstructive Sleep Apnea

Obstructive sleep apnea is a treatable, common condition that is associated with serious health complications and frequently causes excessive daytime sleepiness.[2] In patients with obstructive sleep apnea, structural problems of the upper airway cause obstruction, which results in hypoventilation during sleep. The obstruction may lead to apneic spells, resulting in hypoxemia and hypercapnia. These conditions trigger a sudden adrenergic surge that results in sleep disruption and poor sleep quality. Frequent waking periods lead to excess fatigue and sedation the next day. Risk factors for obstructive sleep apnea include male sex, middle age, elevated BMI, larger-than-average neck circumference, and posterior oropharyngeal crowding on examination. Sleep partners may also report apneic spells and heavy snoring, and patients commonly report headaches that occur in the early morning. Although obstructive sleep apnea is often associated with obesity, it is worth noting that not all patients with this condition are obese, and it can develop in lean patients.[3] Thus, if a clinical suspicion for obstructive sleep apnea is present, diagnostic polysomnography should be performed. Although obstructive sleep apnea cannot entirely be ruled out in this patient's case, it is an unlikely diagnosis given the few features consistent with the condition in his personal or collateral history or on physical examination.

MEDICAL DISORDERS

General medical conditions, including endocrine disorders such as hypothyroidism, adrenal insufficiency, or diabetes mellitus, can cause excessive daytime sleepiness. Fatigue is a prominent feature of these medical conditions. This patient had a normal physical examination, no relevant medical history, and normal laboratory test results. Although results of tests of thyroid function were not provided, he had no symptoms or signs of clinically significant hypothyroidism that would be severe enough to cause his prominent excessive daytime sleepiness.

SUBSTANCE USE

The use of sedating psychotropic agents and ongoing substance use greatly confound this patient's clinical picture, given that both undoubtedly contribute to excessive daytime sleepiness. Ultimately, the venlafaxine dose was reduced, but somnolence persisted. Although opioid use can result in daytime sedation, the urine toxicology test showed only buprenorphine and its metabolites, and no other nonprescribed opioids were detected. However, false negative results can occur owing to dilution, adulteration, or substitution of samples.[4] Of note, the patient reported ketamine use, and ketamine is not measured on traditional toxicology testing. Intoxication with ketamine is associated with impaired consciousness, dissociation, or frank psychosis; however, any effects on sleep and sedation are usually transient.[5] The lack of a clear temporal relationship between the patient's substance use and sedation, as well as the striking periodicity of his symptoms, suggests that substance use is not the cause of his excessive daytime sleepiness.

MAJOR DEPRESSIVE DISORDER

Another possible cause of this patient's excessive daytime sleepiness is major depressive disorder. Although fatigue is a common symptom of depression, it typically leads to poor sleep quality and insomnia. However, some patients with depression have hypersomnia, along with clinically significant fatigue. Depression with hypersomnia is usually associated with worse outcomes, more mixed features, and a higher risk of bipolar disorder. Patients with these conditions should be evaluated for possible bipolar disorder. Among patients with bipolar disorder, hypersomnia is more closely associated with bipolar I disorder than bipolar II disorder.[6,7] This patient's history did not seem to indicate any past episodes of mania or hypomania, but given his age, he may not yet have had a manic episode. Given the early onset and severity of depression, bipolar disorder should remain in the differential diagnosis. However, the likelihood that this condition is the cause of the patient's excessive daytime sleepiness is low.

CENTRAL DISORDERS OF HYPERSOMNOLENCE

The central disorders of hypersomnolence comprise the final category of conditions that can lead to excessive daytime sleepiness.[1,8] These central nervous system disorders are characterized by a disruption of the sleep–wake cycle. Most prominent among these conditions are idiopathic hypersomnia, Kleine–Levin syndrome, and narcolepsy.

Idiopathic Hypersomnia

Idiopathic hypersomnia, as its name suggests, is characterized by excessive daytime sleepiness of unclear cause. It is not related to sleep-onset rapid-eye-movement (REM) periods or REM-related features such as cataplexy, and it manifests as very long sleep periods over the course of 24 hours.

Kleine–Levin Syndrome

Patients with Kleine–Levin syndrome usually have periods of intermittent hypersomnia that can sometimes last for weeks, during which clinically significant cognitive changes occur. Between hypersomnia episodes, patients resume a regular sleep–wake cycle. The patient had no cognitive changes or confusion during periods of hypersomnia, and his symptoms did not appear to have a cyclic pattern; therefore, this diagnosis is unlikely.

Narcolepsy

Narcolepsy is characterized by the sudden onset of REM sleep at any time of the day.[8,9] Sleep episodes in patients with narcolepsy are often irresistible, usually short, frequently associated with dreaming, and typically restore normal wakefulness for up to several hours after an episode. Narcolepsy can be considered a primary or secondary disorder, with the latter resulting from structural damage (e.g., stroke, injury, or mass).

Primary narcolepsy is divided into two types. Narcolepsy type 1 is associated with very low levels of orexin-A in the cerebrospinal fluid (CSF) due to a loss of neurons that produce this neuropeptide in the lateral hypothalamus. It is unclear what causes the loss of neurons, but it is highly associated with specific HLA types, and an underlying autoimmune cause has been suggested.[8,9] Patients with narcolepsy type 1 also commonly have cataplexy, a sudden loss of muscle tone that is most often triggered by strong emotions that are usually positive. Most striated skeletal muscles, with the exception of the diaphragm, are affected, which may lead to sudden collapse without loss of consciousness. Partial manifestations, limited to specific areas such as the face or arms, can lead to a sudden onset of localized weakness. Narcolepsy is also associated with hypnagogic or hypnopompic hallucinations, which refer to hallucinatory phenomena that occur while falling asleep or waking up, as well as sleep paralysis, which is an inability to move while either waking up or falling asleep. Narcolepsy type 2, on the other hand, can often have milder clinical manifestations and is not associated with low orexin-A levels or cataplexy.

This patient had many features that are consistent with narcolepsy type 1. In addition to excessive daytime sleepiness, the patient reported cataplexy, hypnagogic hallucinations, and sleep paralysis. Coexisting depression is commonly seen in patients with narcolepsy, and the patient also had a family history of sleep disorders, which further suggests a genetic component. The presence of cataplexy rules out narcolepsy type 2. Although this patient had two opioid overdoses and we are not aware of whether clinically significant hypoxia of the central nervous system occurred during these episodes, the absence of any other localized neurologic symptoms makes secondary narcolepsy very unlikely.

To establish the diagnosis of narcolepsy type 1 in this patient, I would recommend performing polysomnography, followed by a multiple sleep latency test. I would expect to see an early transition to REM sleep (typically within 15 minutes after sleep onset) on polysomnography and a short time to fall asleep (≤8 minutes) on the multiple sleep latency test.

CLINICAL IMPRESSION

Dr. Quiggle: Initially, the patient's profound fatigue, nonrestorative sleep, and frequent nightmares were attributed primarily to major depressive disorder, anxiety, and active substance use disorder. His symptoms were considered to be associated with psychiatric illness, a disrupted sleep–wake cycle, and medication effects, and it was anticipated that these issues would resolve as his psychiatric condition stabilized.

However, over the course of subsequent visits, the clinical picture evolved. Despite improvement in the patient's mood and the fact that he remained abstinent from substance use, hypersomnolence persisted and became increasingly intrusive, manifesting as unintentional sleep episodes in low-stimulation settings. The severity of his symptoms exceeded what would typically be expected with side effects of antidepressant medications, circadian irregularity, or residual depressive symptoms. In addition, the normal body habitus of the patient, absence of snoring or witnessed apnea, and unremarkable physical examination made obstructive sleep apnea an unlikely diagnosis. He also had no clear trauma history, which argued against other primary psychiatric causes of sleep disruption.

Additional history obtained from the patient included sleep paralysis, hypnagogic hallucinations, and brief laughter-triggered weakness. These features suggested a primary sleep–wake disorder and, together with escalating daytime sleepiness despite psychiatric stabilization and the absence of sedating substances or metabolic derangements, prompted particular concern about narcolepsy type 1. Therefore, the patient was referred to a sleep medicine specialist for further evaluation and confirmatory testing to clarify the underlying sleep disorder and guide management.

CLINICAL DIAGNOSIS

Narcolepsy type 1.

DR. MASOUD P. KAMALI'S DIAGNOSIS

Narcolepsy type 1 with cataplexy.

DIAGNOSTIC TESTING

Dr. Amit Chopra: After a comprehensive clinical sleep evaluation, the patient underwent overnight polysomnography followed by a daytime multiple sleep latency test to assess his excessive daytime sleepiness (Fig. 1). The patient stopped treatment with venlafaxine 2 weeks before the sleep study; however, he continued taking buprenorphine for management of substance use disorder. During polysomnography, sleep-onset latency was 8.2 minutes, and REM-onset latency was 22.0 minutes. The total sleep time was 437.5 minutes, with a sleep efficiency (the percentage of time in bed actually spent sleeping) of 96.4%. The polysomnogram showed no evidence of sleep-disordered breathing, including obstructive and central apnea events, with a score of 0 events per hour on both the apnea–hypopnea index (the total number of episodes of obstructive apnea and hypopnea per hour of sleep) and the respiratory disturbance index (the number of episodes of obstructive apnea, hypopnea, and respiratory-effort–related arousal per hour of sleep). In addition, there was no evidence of sleep-related hypoxemia or hypoventilation, and no excessive periodic limb movements or parasomnia behaviors were noted. The results of the multiple sleep latency test were consistent with a diagnosis

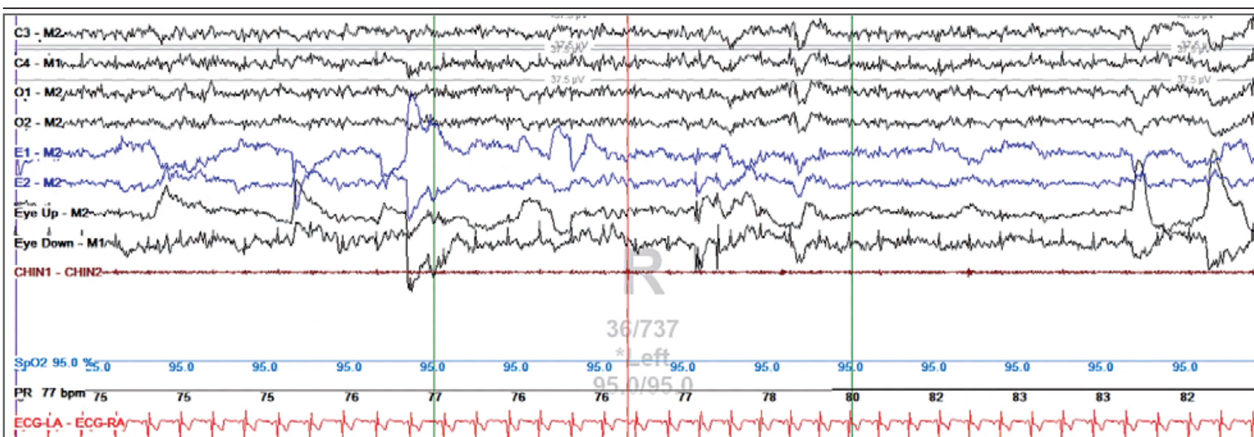

Figure 1. Polysomnogram and Multiple Sleep Latency Test.

Shown is a 30-second sleep segment, which includes data for the following electrophysiological channels: electroencephalogram (central leads, C3–M2 and C4–M1; occipital leads, O1–M2 and O2–M2), chin electromyogram, electro-oculogram (EOG; leads, E1–M2 and E2–M2), electrocardiogram (ECG; leads, LA [left arm] and RA [right arm]), pulse rate (PR), and oxygen saturation (SpO_2). The sleep-onset rapid-eye-movement (REM) period is characterized by conjugate rapid eye movements in the EOG channels and REM sleep with atonia in the chin electromyogram channel.

of a central disorder of hypersomnolence, with a mean sleep-onset latency of 1.6 minutes (sleep was observed during four of four naps) and three sleep-onset REM periods. These findings met the criteria for narcolepsy type 1 (a mean sleep-onset latency of ≤8 minutes and at least two sleep-onset REM periods), and, along with clinical symptoms of cataplexy, established the diagnosis in this patient.

SLEEP MEDICINE MANAGEMENT

Dr. Chopra: Narcolepsy type 1 is a central disorder of hypersomnolence characterized by cataplexy or by very low levels of orexin-A (<110 pg per milliliter) in CSF obtained by lumbar puncture.[10] The prevalence of narcolepsy type 1 is estimated to be 1 case per 2000 persons in the United States and Europe.[11,12] It is characterized by a selective loss of orexin-A–producing neurons in the lateral hypothalamus,[13,14] which is most likely due to an autoimmune process.[15] These neurons help sustain prolonged wakefulness and trigger REM sleep through production of orexin-A, which regulates the sleep–wake cycle and also modulates other brain functions, including the reward system and metabolism, by acting on histamine and other monoamine-producing neurons.[16]

Narcolepsy type 1 is often undiagnosed or misdiagnosed, with a mean interval between symptom onset and diagnosis of approximately 8 to 10 years.[17] This patient had a pentad of symptoms suggestive of narcolepsy type 1, including excessive daytime sleepiness, cataplexy, sleep-related hallucinations, sleep paralysis, and disturbed night sleep (Fig. 2). The patient was noted to have coexisting major depressive disorder and substance use disorder, both of which substan-

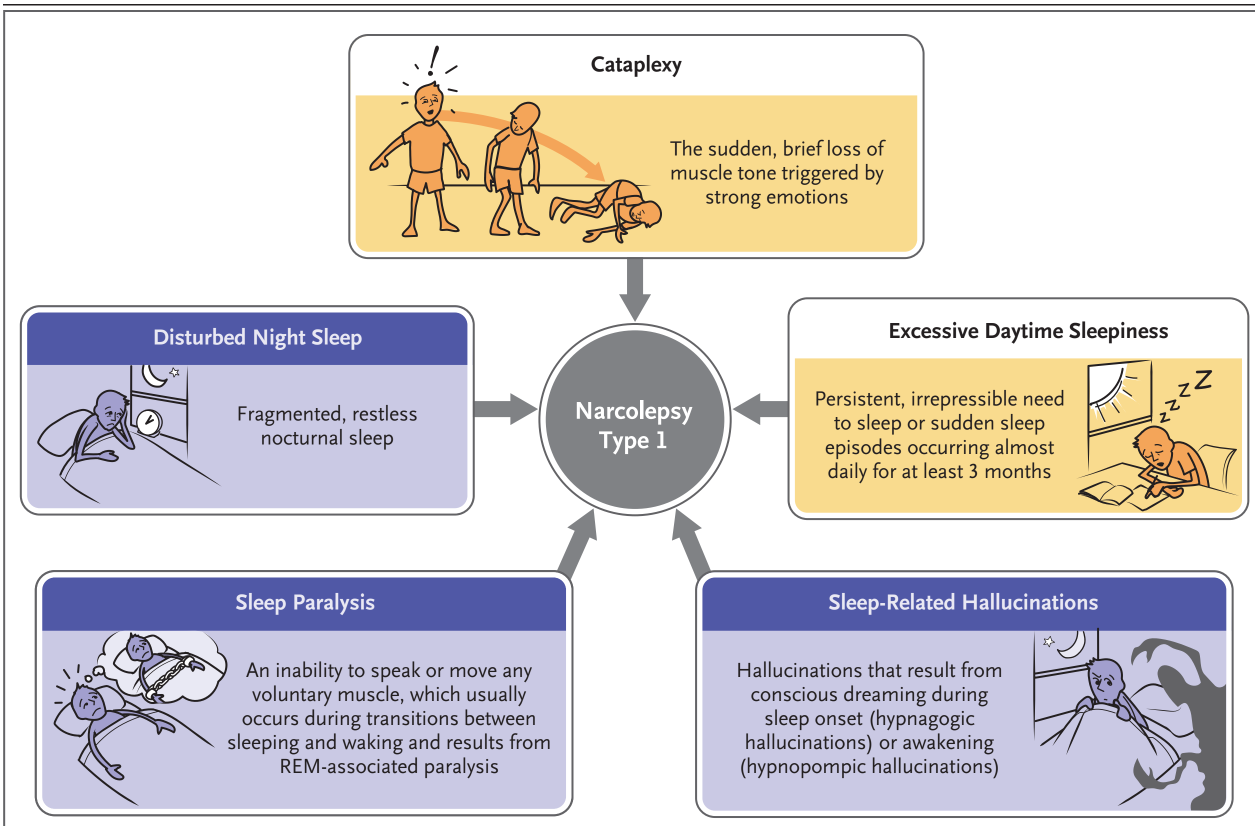

Figure 2. Pentad of Symptoms of Narcolepsy Type 1.
The main symptoms of narcolepsy type 1 are cataplexy, excessive daytime sleepiness, disturbed night sleep, sleep paralysis, and sleep-related hallucinations.

tially affected his daily functioning. It is noteworthy that psychiatric disorders, including depression, anxiety, and attention deficit–hyperactivity disorder, are more prevalent among patients with narcolepsy than in the general population.[18] In this case, the symptoms of excessive daytime sleepiness preceded the onset of the patient's mood disorder and substance use disorder and persisted during periods of euthymia and abstinence from substance use. Diagnostic sleep studies, including polysomnography followed by the multiple sleep latency test, confirmed the diagnosis of narcolepsy type 1, since he had a mean sleep-onset latency of 1.6 minutes across four naps and three sleep-onset REM periods during the multiple sleep latency test, along with a clinical history of cataplexy.

Effective management of narcolepsy type 1 involves both nonpharmacologic and pharmacologic interventions.[16] Nonpharmacologic treatment strategies include self-care, behavioral therapy, scheduled napping and regular night sleep, self-help groups, and psychotherapy.[19] Regular physical activity, appropriate caffeine use, and a balanced diet that limits carbohydrate intake are also recommended.[16] Current pharmacologic treatments to manage narcolepsy type 1 target wake-promoting neurotransmitters, including dopamine, norepinephrine, and histamine. The first-line treatments for excessive daytime sleepiness in patients with narcolepsy type 1 include wake-promoting agents, such as modafinil, armodafinil, and solriamfetol. In contrast, sodium oxybate and pitolisant are indicated as first-line treatments for excessive daytime sleepiness and cataplexy.[16,20] The second-line treatments for excessive daytime sleepiness include methylphenidate and amphetamine salts. Antidepressants such as selective serotonin-reuptake inhibitors, selective serotonin–norepinephrine-reuptake inhibitors, and tricyclic antidepressants are considered to be second-line agents for the management of cataplexy.[16,20]

After the diagnosis was confirmed, the patient received a prescription for modafinil for the management of excessive daytime sleepiness associated with narcolepsy type 1.

FOLLOW-UP

Dr. Quiggle: Within several weeks after starting treatment with modafinil, the patient reported a marked reduction in daytime fatigue and an improved ability to sustain wakefulness during routine activities. His total sleep time normalized, and the frequency of sleep paralysis and hypnagogic hallucinations markedly diminished. He described increased concentration and engagement in daily tasks and noted a renewed capacity to participate in work and academic planning.

The patient characterized the change as transformative, stating, "I feel like I have another chance at life" and "this is how life could have been all along." His symptoms of social anxiety persist, and he continues to receive treatment with escitalopram and weekly psychotherapy.

FINAL DIAGNOSIS

Narcolepsy type 1.

This case was presented at Psychiatry Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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